PAYTON'S LiDAR SENSOR
LASER SPEED · TF03 LiDAR · DESIGNED BY PAYTON PITTS

Sprint report
TF03 LiDAR · 1000 Hz · **±0.03 s · ±0.1 m/s**

MAX VELOCITY · 30 M SPRINT

Demo Athlete

Nov 8, 2026 · 7:50 PM Body weight **68.0 kg** TF03 LiDAR · 1000 Hz · ±0.03 s · ±0.1 m/s

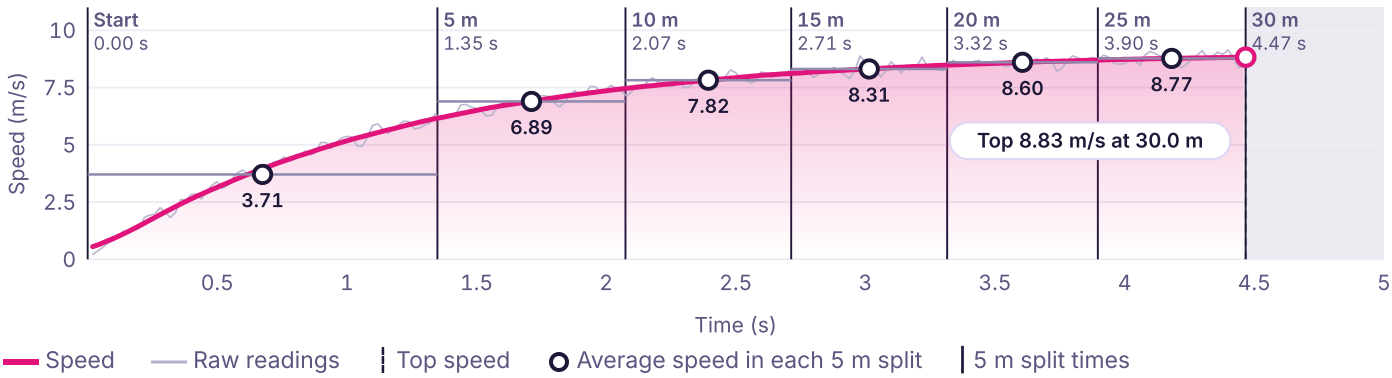
30 m time	Top speed	Distance to top speed
4.47 s	**8.83** m/s	**30.0** m
Start triggered at 0.200 m/s	19.7 mph · fastest stride	95% of top by 18.5 m
	Fastest flying 10 m	**Power**
	1.15 s	**1186** W
	20.0–30.0 m · 8.69 m/s	17.4 W/kg · peak horizontal

Summary · written by the app from this run

- Demo reached a top speed of **8.83 m/s** (19.7 mph) at 30.0 m.
- At top speed: about **4.38 steps/s** and **2.02 m** per step. 98% of top speed by 23.6 m.
- The fastest flying 10 m was **1.15 s**, from 20.0 to 30.0 m (8.69 m/s).
- Best last 5 m on the runway is 8.16 m/s, 92% of this sprint top speed.

NEXT SESSION Top speed came at 30.0 m, right at the line, so it is probably still rising. Test a longer run (40 m) or a flying 20 m with a longer build-up to find the real max.

Speed over time



Splits

SEGMENT	AVERAGE SPEED	TIME
0–5 m	3.71 m/s	1.35 s
5–10 m	6.89 m/s	0.73 s
10–15 m	7.82 m/s	0.64 s
15–20 m	8.31 m/s	0.60 s
20–25 m	8.60 m/s	0.58 s
25–30 m	8.77 m/s	0.57 s

Flying 10 m (20.0–30.0 m) **1.15 s**

0–10 m **2.07 s**

0–20 m **3.32 s**

0–30 m **4.47 s**

Top-speed profile

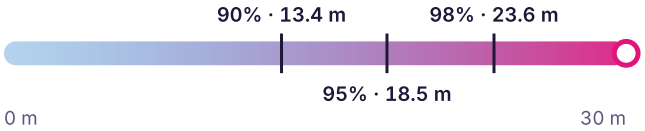
Top speed **8.83 m/s** (19.7 mph)

Step rate **4.38 /s** steps per second

Step length **2.02 m** per step

Step rate and length are estimates, taken at top speed. Top speed = step rate × step length (4.38 × 2.02 = 8.83 m/s). When top speed goes up, check which of the two moved.

SPEED BUILD-UP: WHERE YOU HIT 90 / 95 / 98% OF TOP SPEED



Ceiling (speed the curve was heading for)	9.00 m/s
Top speed as % of ceiling	98%
Best last 5 m on the runway	8.16 m/s
Runway speed as % of sprint top speed	92%

What these numbers mean for your top speed

Top speed
The fastest single stride in the run. In the vault this is the number that matters most: the faster you can run, the faster you can arrive at the plant. Test it every 2–3 weeks with full rest between reps, and follow the trend in Analytics.

Distance to top speed
How much run-up you need to reach your max. If you hit 98% by 25 m, a shorter approach means you plant before you're at full speed. The fixed 100 ft and 150 ft setups show how much of this speed you keep on the runway with a pole.

Ceiling
The top speed the speed curve was heading toward. If your top speed is under about 97% of the ceiling, the test was too short to show your real max. Run a longer one.

Fastest flying 10 m
The steadiest way to test max speed, since it doesn't depend on the start. Build up for 20–30 m, then sprint 10 m. Take the best of 2–3 reps. Every 0.01 s off a flying 10 m is about 0.07–0.08 m/s more speed.

Step rate × step length
The only two ways to run faster. Watch which one rises when top speed improves. Step length comes from force into the ground at speed (strength, bounding, stiff ankles). Step rate comes from quick ground contacts and a fast leg swing.

Runway %
How much of your sprint speed you carry into the last 5 m with the pole. Around 90–95% is a common range. If it's low, the pole carry and run-up rhythm are costing you speed, not your sprinting.

All parameters

PARAMETER	VALUE	ALSO
MAX VELOCITY		
Top speed (fastest stride)	8.83 m/s	19.7 mph
Distance to top speed	30.0 m	–
Time to top speed	4.47 s	–
Fastest flying 10 m: time	1.15 s	–
Fastest flying 10 m: speed	8.69 m/s	–
Fastest flying 10 m: from	20.0 m	–
Reached 90 / 95 / 98%	13.4 / 18.5 / 23.6 m	–
Step rate at top speed	4.38 steps/s	estimate
Step length at top speed	2.02 m	estimate
Ceiling (V0)	9.00 m/s	98% reached
TEST		
30 m time	4.47 s	–
Average speed	6.72 m/s	–
Starting threshold	0.200 m/s	–
Body weight	68.0 kg	–
Readings per second	1000	–

PARAMETER	VALUE	ALSO
POWER AND FORCE		
Power (P0)	1186 W	17.4 W/kg
Peak horizontal power	1186 W	17.4 W/kg
Mean horizontal power	589 W	8.7 W/kg
Start force (F0)	527 N	7.75 N/kg
Force–velocity slope	-0.8621	–
ACCELERATION (LESS IMPORTANT FOR VAULT)		
Peak acceleration	7.26 m/s²	–
Mean acceleration to top speed	1.98 m/s²	–
SPLITS		
0–5 m	1.35 s	3.71 m/s
5–10 m	0.73 s	6.89 m/s
10–15 m	0.64 s	7.82 m/s
15–20 m	0.60 s	8.31 m/s
20–25 m	0.58 s	8.60 m/s
25–30 m	0.57 s	8.77 m/s

How accurate is this?

Times: ±0.03 s

The TF03 measures distance to within about 10 cm up close and 1% farther out, 1000 times a second. At sprint speed, 10 cm is about 0.01 s. Times start when speed passes 0.2 m/s, so the reaction to a start signal isn't included.

Keep it consistent

Same tripod height (lower back), aimed straight down the lane, same surface and shoes. Compare tests run the same way.

Checked so far

These limits come from the sensor's spec and from simulated runs. Check them against timing gates or video once real runs are recorded.

Speeds: ±0.1 m/s

Speed comes from one smoothed distance line, averaged over each stride, so a single noisy reading can't set a top speed. Splits and the speed curve come from the same line. Step rate and length are estimates from the small speed dip at each foot contact.

Power

Peak horizontal push, which comes mostly from the first few steps. Acceleration doesn't decide a vault approach, so use it only to check that strength work is carrying over to the track.

PAYTON'S LiDAR SENSOR
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